(T)(J-24) Roll No. 1101294

PAPER ID—16148

B. Tech. EXAMINATION, 2024

(First Semester)

ELECTRONICS AND COMMUNICATION ENGINEERING

Code : BSP-101

Physics

Time : 3 Hours *Maximum Marks : 70*

Before answering the question-paper candidates should ensure that they have been supplied to correct and complete question-paper. No complaint, in this regard, will be entertained after the examination.

Note : Unit I has seven questions. All the questions are compulsory. All the questions carry 2 marks. Unit II has six questions. *Four* questions need to be attempted from this Unit. All the questions has two parts of 7 marks each.

Unit I

1. Which types of statistics shall be applicable for a gas of photons ? Justify your answers. 2

2. Discuss the properties of the wave function. 2

3. Draw the Fermi level in *p*-type and *n*-type semiconductors. 2

4. Differentiate between spontaneous emission and stimulated emission. 2

5. Explain Photovoltaic effect. 2

6. Determine the de-Broglie wavelength for an electron moving with a speed of 10^7 m/s. Mass of the electron is 9.1×10^{-31} kg $\hbar = 6.6 \times 10^{-34}$ J-s. 2

7. What is Fermi's Golden rule ? 2

Unit II

1. (a) Derive Schrödinger time dependent wave equation for a free particle. Explain all terms. 7

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(b) How did the motion of electrons in periodic potential lead to the concepts of energy bands and effective mass ? Explain. Draw E-K diagram and discuss the salient characteristics. 7

2. (a) Explain the concept of Fermi level and Fermi energy. In an *n*-type semiconductor, the Fermi level lies 0.3 eV below the conduction band at 300 K. If the temperature is increased to 330 K. Find the new position of the Fermi level. 7

(b) The intrinsic resistivity of germanium at 300 K is 0.47 ohm-m. The electron mobility μ_e at 300 K in germanium is 0.39 m^2/V-s. The hole mobility μ_h at 300 K in germanium is 0.19 m^2/V-s. Calculate the density of electrons in the intrinsic material. Also calculate the drift velocity of holes and electrons for an electric field $E = 10^4$ V/m. 7

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3. (a) What is Hall Effect ? Describe an experimental set-up for the measurement of Hall voltage and Hall coefficient. Explain, how Hall coefficient is used to determine the mobility of charge carriers. 7

(b) Draw the energy band diagram of an unbiased *p-n* junction. Explain the terms : Barrier energy, barrier potential, and depletion region. 7

4. (a) Discuss the working principle of Deep Level Transient Spectroscopy (DLTS). 7

(b) What is the difference between four point probe method and van der Pauw method for resistivity measurement ? 7

5. (a) What do you understand by quantum confinement ? Discuss density of states for different types of quantum confinements. 7

(b) Describe popular techniques used for preparation and characterization of quantum dots. 7

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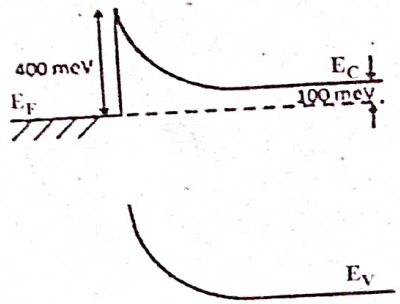
6. (a) An ideal diode is described by $I_D = I_0 \{\exp(qV_A/k_BT) - 1\}$ but real diodes have series resistance Consider a diode with $I_0 = 10^{-12}$ A at room temperature. A forward bias of 1 V is applied and a current of 1 A is measured. What is the series resistance of the diode ? 7

(b) Consider an ideal metal-semiconductor junction with the band diagram shown below. The semiconductor has a relative dielectric constant of 12 and an electron affinity of 4.0 eV. Answer the following questions :



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(i) What is the work function of the metal ?

(ii) What is the built-in potential of the semiconductor ?

(iii) At V = 0, we measure a capacitance of 17 pF for a 100 μm × 100 μm diode. What is the doping density in the semiconductor ? 7